

REGION 3

Set up all six orders of integration for

$$\iiint_{R_3} f(x, y, z) dV,$$

where R_3 is the region bounded by

$$y = x, \quad y = \sqrt{2 - x^2}, \quad z = 2x^2 + y^2, \quad z = 4 - y^2, \quad \text{and} \quad 0 \leq x \leq 1.$$

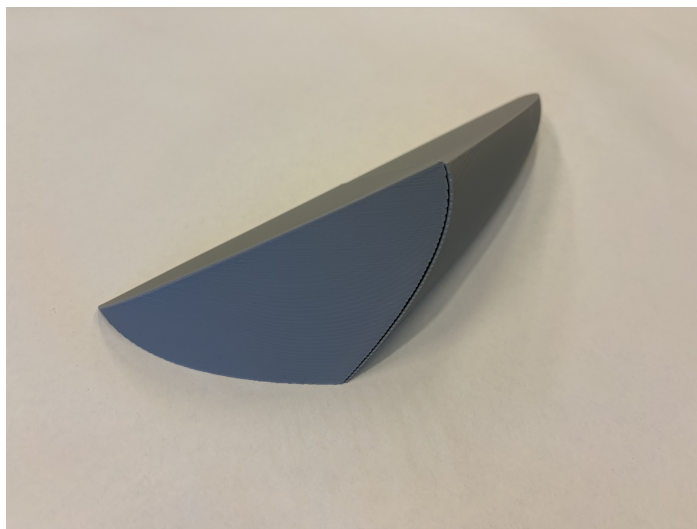
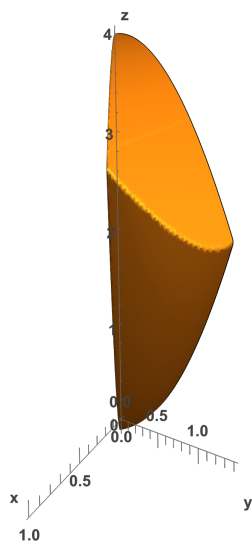


FIGURE 1. Computer-modeled and 3D-printed views of R_3

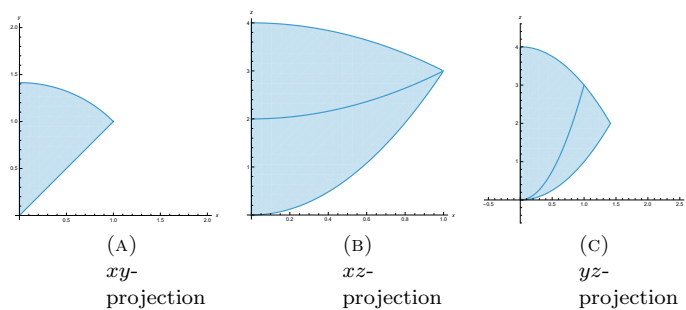


FIGURE 2. Projections of R_3 into the three coordinate planes.

The following triple integrals represents integrating over the region using the xy -projection:

$$\int_1^{\sqrt{2}} \int_0^{\sqrt{2-y^2}} \int_{2x^2+y^2}^{4-y^2} f(x, y, z) dz dx dy + \int_0^1 \int_0^y \int_{2x^2+y^2}^{4-y^2} f(x, y, z) dz dx dy$$

or

$$\int_0^1 \int_x^{\sqrt{2-x^2}} \int_{2x^2+y^2}^{4-y^2} f(x, y, z) dz dy dx$$

The following triple integrals represent integrating over the region using the xz -projection:

$$\int_3^4 \int_0^{\sqrt{4-z}} \int_x^{\sqrt{4-z}} f(x, y, z) dy dx dz + \int_2^3 \int_0^{\sqrt{z-2}} \int_x^{\sqrt{4-z}} f(x, y, z) dy dx dz$$

$$+ \int_2^3 \int_{\sqrt{z-2}}^{\sqrt{\frac{z}{3}}} \int_x^{\sqrt{z-2x^2}} f(x, y, z) dy dx dz + \int_0^2 \int_0^{\sqrt{\frac{z}{3}}} \int_x^{\sqrt{z-2x^2}} f(x, y, z) dy dx dz$$

or

$$\int_0^1 \int_{x^2+2}^{4-x^2} \int_x^{\sqrt{4-z}} f(x, y, z) dy dz dx + \int_0^1 \int_{3x^2}^{x^2+2} \int_x^{\sqrt{z-2x^2}} f(x, y, z) dy dz dx$$

The following triple integrals represent integrating over the region using the yz -projection:

$$\int_3^4 \int_0^{\sqrt{4-z}} \int_0^y f(x, y, z) dx dy dz + \int_0^3 \int_0^{\sqrt{\frac{z}{3}}} \int_0^y f(x, y, z) dx dy dz$$

$$+ \int_2^3 \int_{\sqrt{\frac{z}{3}}}^{\sqrt{4-z}} \int_0^{\sqrt{\frac{1}{2}(z-y^2)}} f(x, y, z) dx dy dz + \int_0^2 \int_{\sqrt{\frac{z}{3}}}^{\sqrt{z}} \int_0^{\sqrt{\frac{1}{2}(z-y^2)}} f(x, y, z) dx dy dz$$

or

$$\int_0^1 \int_{3y^2}^{4-y^2} \int_0^y f(x, y, z) dx dz dy + \int_0^1 \int_{y^2}^{3y^2} \int_0^{\sqrt{\frac{1}{2}(z-y^2)}} f(x, y, z) dx dz dy + \int_1^{\sqrt{2}} \int_{y^2}^{4-y^2} \int_0^{\sqrt{\frac{1}{2}(z-y^2)}} f(x, y, z) dx dz dy$$

All of these triple integrals evaluate to $\frac{\pi}{2}$ when $f(x, y, z) = 1$.